

# A Language-Agnostic Framework for Dependency Graph Construction

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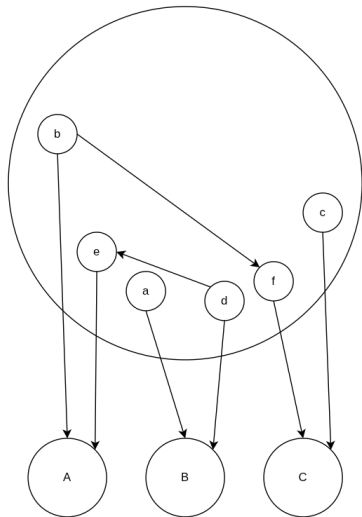
# Assessing Software Quality

## Definition

An **architectural smell** is a sign of poor design choices on the architectural level.

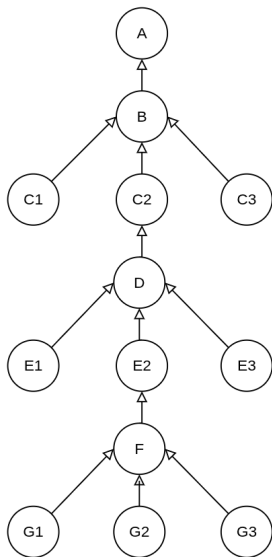
- Similar to the more known Code Smells.
- They can be detected by analyzing code shape, components and the dependency graph.
- The presence of many smells in a software project increases the estimate of **Technical Debt**, which indicates the future cost of maintenance, development and evolution.
- By Lehman's laws of software evolution we can expect it to grow over time, as bad practises effects stratify

# God Component



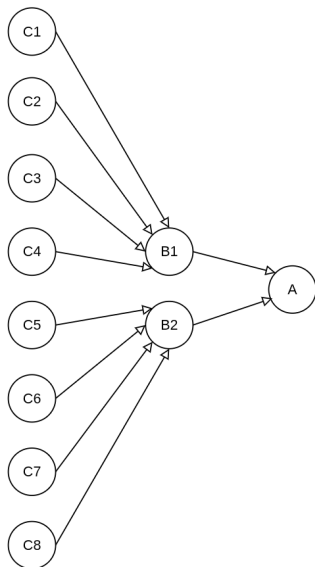
- One big component with a lot of heterogeneous sub-components inside
- Probably takes care of a lot of different concerns
- Probably has a lot of outwards dependencies and a few inwards dependencies
- Probably is a consistent percentage of the code
- Typical of Legacy Systems (especially **monolithic** architectures)

# Deep Hierarchy



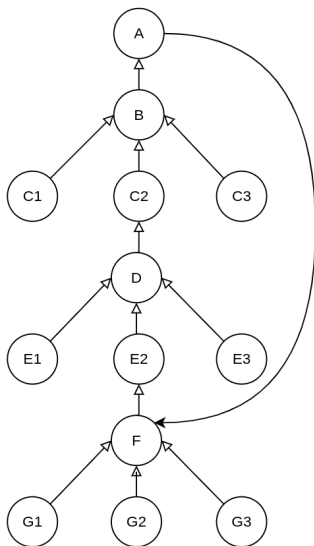
- A long inheritance chain, each level adding a bit of behaviour
- Suggests high implementation sparsity
- Suggests abuse/loss of generalization
- Typical of Object-Oriented Programming

# Wide Hierarchy



- A hierarchy chain with at each level many children
- Suggests high implementation sparsity
- Suggests abuse/loss of generalization
- Typical of Object-Oriented Programming

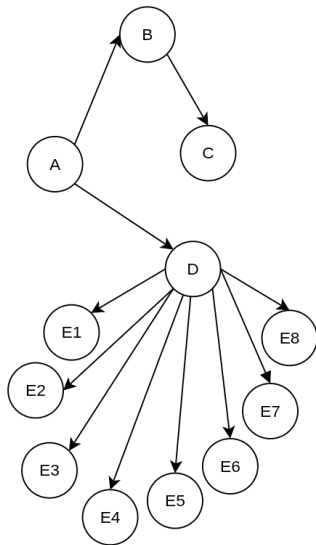
# Cyclic Hierarchy



- Like the previous, but with a twist
- A parent component is actively depending on a child component
- Suggests a short-circuit of abstractions
- Typical of Object-Oriented Programming

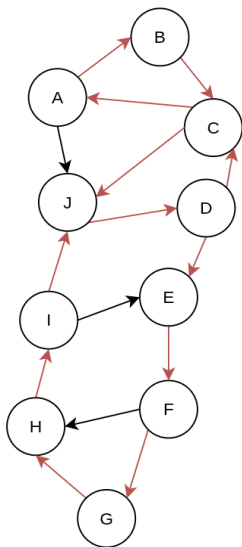


# Unstable Dependency



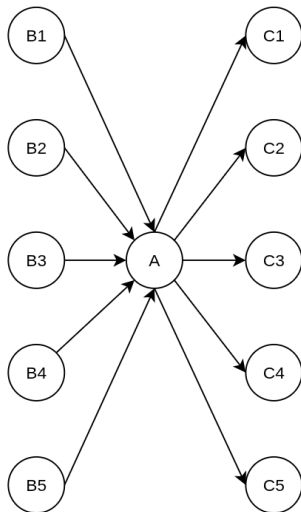
- A stable component depending on a unstable component
- (In)stability is usually defined by the ratio  $\frac{C_e}{C_a + C_e}$
- Such a component is difficult to maintain stable because its dependencies have a lot of reasons to change for

# Cyclic Dependency



- Components involved in a cycle are difficult to change
- Typical of legacy systems

# Hub-Like Dependency



- A component with a lot of inwards and outwards dependencies
- Too stable to allow for changes
- Too unstable to depend on
- Typical of many legacy library components



JUnit4

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OVERVIEWASSESSMENTCOMPREHENSION

Architectural smells

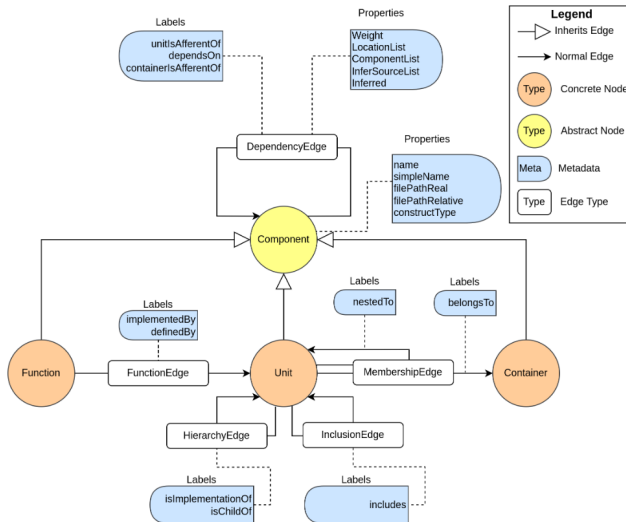




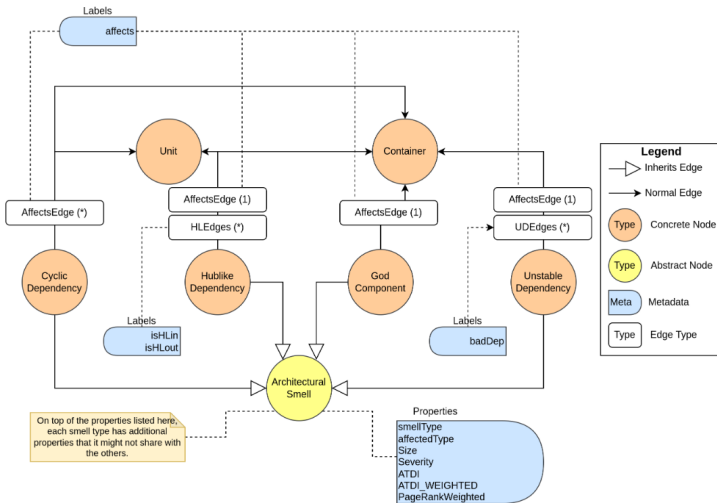


| ID      | Smell Type         | TechDebt Index | Severity | Affected Type | Affected Elements             | Size |
|---------|--------------------|----------------|----------|---------------|-------------------------------|------|
| > 11606 | Hublike Dependency | 300            | 8.00     | Package       | org.junit.runner              | 21   |
| > 11324 | Cyclic Dependency  | 107            | 5.00     | Package       | org.junit.experimental.the... | 2    |
| > 11174 | Cyclic Dependency  | 99             | 6.00     | Package       | org.junit.runner, org.juni... | 8    |
| > 11232 | Cyclic Dependency  | 95             | 6.00     | Package       | org.junit.internal.runners... | 8    |
| > 11123 | Cyclic Dependency  | 94             | 6.00     | Package       | org.junit.runner, org.juni... | 6    |
| > 11244 | Cyclic Dependency  | 93             | 5.00     | Package       | org.junit.internal.runners... | 7    |
| > 11267 | Cyclic Dependency  | 91             | 6.00     | Package       | org.junit.internal.runners... | 8    |

# The Dependency Graph



# The Architectural Smell Graph



# The "Tower of Babel" Problem

# Language Segmentation

Computing language usage shares is not easy ...

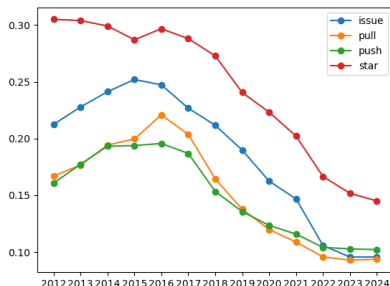


Figure: Stats of JavaScript

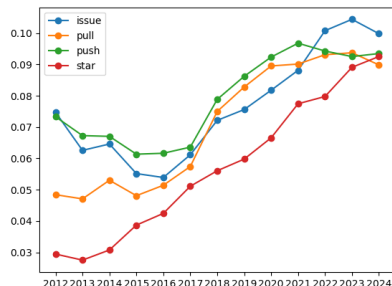
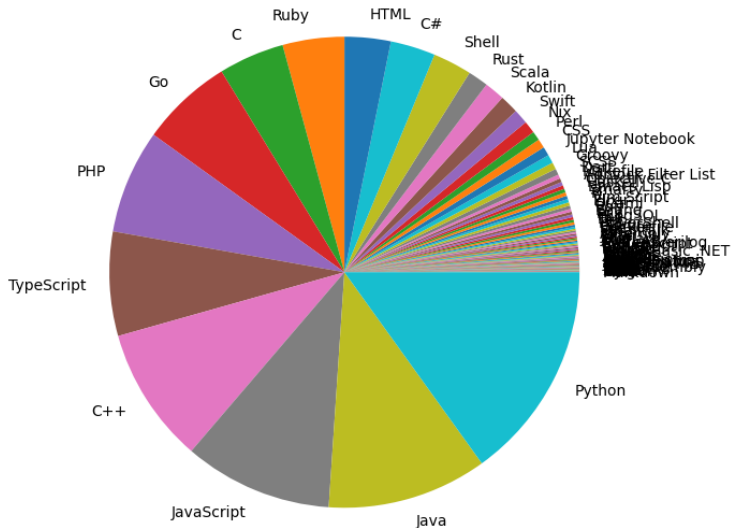


Figure: Stats of C++



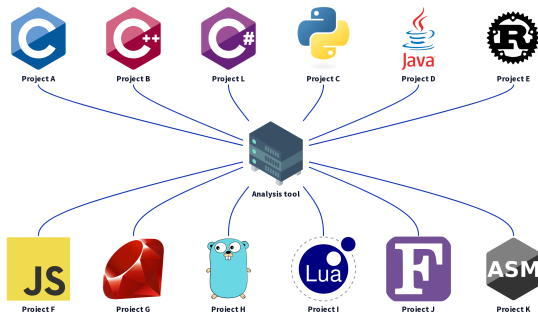
# Language Segmentation

... but we can do some estimates



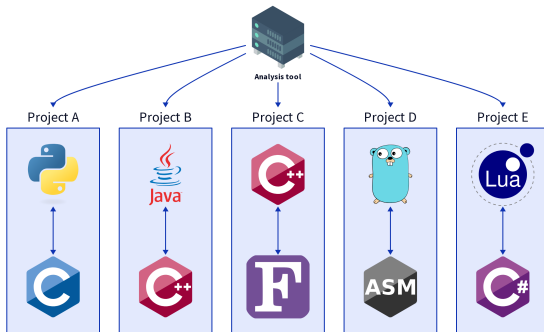
# A matter of scale

- A software analysis tool need to work with many programming languages.
- Each deployed instance need every language interface to be useful
- Each language interface is a reason to change and increases maintenance costs



# A matter of perspective

- A project can use more than one language, which is usually not supported by tools
- Each language interface usually need a different implementation for the same logic
- Sometimes it is easier to develop a distinct version of each language



# A Bit of History

# Language Frontends

- Common approach: use language frontends (e.g., Clang, Eclipse JDT/CDT) to parse code and extract data.
- APIs are heterogeneous and lack standardization.
- The *Language Server Protocol (LSP)* aims to unify such interfaces, but still has limits in query expressiveness and flexibility.
- Frontends are designed for IDEs/compiler, requiring extra adaptation for analysis tools.
- Typically heavy-weight — include many elaborations not needed by analysis tools.

# Manual/Automatic Transpilation

- Some works avoid using multiple language frontends by translating source code into a single target language or DSL.
- Translation can be done manually or via transpilers.
- Works well for languages designed for transpilation (e.g., TypeScript, Haxe).
- For other languages, translation may lose semantic details (e.g., classes, namespaces, packages).
- To preserve semantics, the target language must be as expressive as the source.
- This approach remains inefficient:
  - Automatic translation between languages is still an open research problem.
  - Manual translation is time-consuming and error-prone.
  - Translated code must still be parsed and analyzed afterward.

# AST/CST Translation

- Another approach: translate a **language-dependent AST** into a **language-independent AST**.
- Enables analysis tools to apply algorithms on a common, unified representation.
- Can also be applied to **Concrete Syntax Trees (CSTs)**, which capture full program syntax.
- Typically relies on language frontends (manual or generated) to perform near 1:1 translation into an *extended AST (eAST)*.
- Sometimes produces an intermediate serialized form (e.g., XML).

- A **meta model abstraction** offers a language-agnostic alternative:
  - Represents high-level program entities — classes, structs, functions — instead of raw syntax trees.
  - Simplifies cross-language analysis by focusing on shared structural concepts.
  - Reduces dependency on language frontends and heavy parsing.
- Enables more flexible and extensible analysis frameworks built around **semantic equivalence** rather than syntax.
- Often requires custom frontends to build the meta-model from source code.

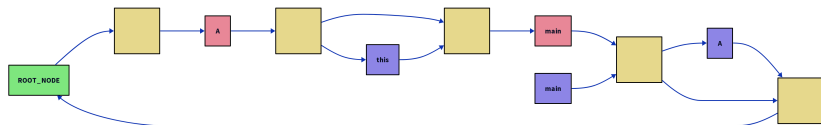


- Beyond traditional frontends, new data structures have been designed for specific analysis tasks (e.g., code navigation, reference resolution).
- Evaluated the use of **Stack Graphs** — composable graphs representing identifiers and their relationships — for language-independent reference resolution.
- Showed promising accuracy but faced:
  - Practical and scalability limitations.
  - Partial language-independence: graph construction still required language-specific CST queries.
- Despite limitations, the study highlighted:
  - The potential of **language-independent analysis**.
  - Its value as a guiding principle for future software analysis strategies.

# ADS / Stack Graphs

- Composable graph
- Represents identifiers and scopes of source code
- Enables language agnostic reference resolution

```
class A {  
    public static void main() {  
    }  
}
```



# ADS / Tree Sitter Stack Graphs

- A "TSSG" Grammar is composed of pairs
- Each pair matches a section of the CST with some procedural code
- Definition of nodes, edges and labels

```
(field_access object: (_)@object field: (_)@field)@this {  
  node @this.scope  
  node @this.pop_start  
  node @this.pop_end  
  edge @field.push_end -> @object.push_start  
  let @this.push_start = @field.push_start  
  let @this.push_end = @object.push_end  
  edge @this.push_end -> @this.scope  
  edge @this.pop_start -> @this.push_start  
  attr (@field.push_start)  
    is_reference, refkind = "accessField"  
}
```

# A New Solution

# Very different ...

```
import numpy

def dct(X: numpy.ndarray) -> numpy.ndarray:
    if len(X.shape) == 1:
        M = X.shape[0]
        return dct(X.reshape((1, M)).reshape((M,)))

    N, M = X.shape
    Y: numpy.ndarray = numpy.zeros((N, M), dtype=numpy.float64)

    pi_over_M = numpy.pi / M
    sqrt_of_2_over_M = numpy.sqrt(2.0 / M)
    one_over_sqrt_of_two = numpy.sqrt(0.5)
    j_minus_half_from_1_to_M = (numpy.array(range(M)) + 0.5)

    for i in range(0, N):
        for k in range(0, M):
            k_pi_over_M = k * pi_over_M
            Y[i, k] = sum(
                numpy.cos(
                    k_pi_over_M *
                    j_minus_half_from_1_to_M *
                    X[i]) * sqrt_of_2_over_M
            )
            Y[i, 0] = Y[i, 0] * one_over_sqrt_of_two
        return Y

def dct2(X: numpy.ndarray) -> numpy.ndarray:
    return numpy.transpose(dct(numpy.transpose(dct(X))))
```

```
#include <dcct/slow_actuator.hh>

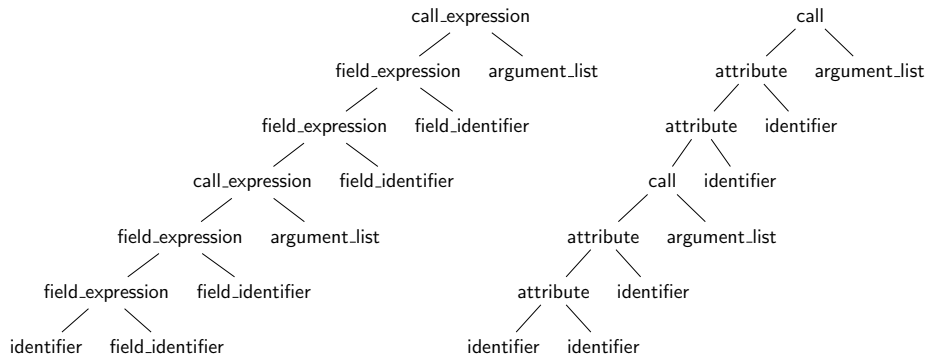
Eigen::MatrixXd dcct::SlowActuator::dct(Eigen::MatrixXd& X) {
    uint32_t N = X.rows(), M = X.cols();
    Eigen::MatrixXd Y(N, M);

    double_t pi_over_M = std::numbers::pi / (double_t) M;
    double_t sqrt_of_two_over_M = std::sqrt(2.0 / (double_t) M);
    double_t one_over_sqrt_of_two = std::sqrt(0.5);

    for (uint32_t i = 0; i < N; ++i) {
        for (uint32_t k = 0; k < M; ++k) {
            double_t sum = 0;
            double_t _k_pi_over_M = pi_over_M * k;
            for (uint32_t j = 0; j < M; ++j) {
                sum += std::cos(_k_pi_over_M * (j + 0.5)) * X.coeff(i, j);
            }
            Y.coeffRef(i, k) = sum * sqrt_of_two_over_M;
        }
        Y.coeffRef(i, 0) *= one_over_sqrt_of_two;
    }
    return Y;
}

Eigen::MatrixXd dcct::SlowActuator::dct2(Eigen::MatrixXd& X) {
    Eigen::MatrixXd Y = dct(X).transpose();
    X = dct(Y).transpose();
    return X;
}
```

## ... Or very similar?



# Tree Sitter



- General enough to parse any programming language
  - Fast enough to parse on every keystroke in a text editor
  - Robust enough to provide useful results even in the presence of syntax errors
  - Dependency-free so that the runtime library (which is written in pure C11) can be embedded in any application
- Tree-sitter is a parser generator tool and an incremental parsing library.
  - It can build a concrete syntax tree for a source file and efficiently update the syntax tree as the source file is edited.

# Why Tree Sitter?

Tree Sitter is not only easy to operate, but it is also easy to implement parsers for new languages.

```
typedef [0, 0] - [0, 32]
  name: identifier [0, 8] - [0, 12]
  type: integer_type [0, 15] - [0, 31]
    size: integer [0, 23] - [0, 24]
    signed: boolean [0, 26] - [0, 30]
function [1, 0] - [1, 32]
  name: identifier [1, 3] - [1, 9]
  parameters: parameter_list [1, 9] - [1, 31]
    parameter [1, 10] - [1, 20]
      name: identifier [1, 10] - [1, 13]
      type: pointer_type [1, 15] - [1, 20]
        type: identifier [1, 16] - [1, 20]
    parameter [1, 22] - [1, 30]
      name: identifier [1, 22] - [1, 23]
      type: pointer_type [1, 25] - [1, 30]
        type: identifier [1, 26] - [1, 30]
```

Figure: Example of Lart CST

```
typedef char = integer<8, true>;
fn printf(fmt: &char, s: &char);
```

Figure: Example of Lart code

```
function: $ => seq(
  'fn',
  field('name', $.identifier),
  field('parameters', $.parameter_list),
  optional(seq('>', field('type', $_.type))),
  choice(field('body', $.block), ';')
),

include: $ => seq(
  'include',
  choice(
    seq('<', field('globalpath', $.path_literal), '>'),
    seq('"', field('localpath', $.path_literal), '"')
  ), ';'
),
```

Figure: Slice of Lart grammar as TS config



# The Approach

## Development

- Gain initial knowledge of a language's **grammar**
- Gain initial knowledge of a language's **semantics**
- Prepare a **language benchmark** to verify that the implementation is correct
- Each programming language shall be described as a **feature set**

## Operation

- Extract CSTs with **Tree Sitter** for each file
- Traverse the CSTs to capture the components definitions and the references
- Construct a *Language-Independent* **Component Tree**
- Accumulate the **Dependency Queries**
- Resolve the Dependency Queries with the Component Tree into the **Dependency Graph**
- Export the Dependency Graph to JSON

# How it works / CST Traversal / Analysis

## Java

```
(method_declaration
  type: (void_type)
  name: (identifier)
  parameters: (formal_parameters)
  body: (block))
```

## Rust

```
(function_item
  name: (identifier)
  parameters: (parameters)
  return_type: (unit_type)
  body: (block))
```

## C++

```
(function_definition
  type: (primitive_type)
  declarator: (function_declarator
    declarator: (identifier)
    parameters: (parameter_list))
  body: (compound_statement))
```

## Python

```
(function_definition
  name: (identifier)
  parameters: (parameters)
  return_type: (type (none))
  body: (block (pass_statement)))
```

# How it works / CST Traversal / Translation

## Java

```
if tsnode.type == "method_declaration" {  
  let function = Function {  
    name: process_name(tsnode.get_field("name")),  
    type: process_type(tsnode.get_field("type")),  
  }  
  process_parameters(tsnode.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

## Rust

```
if tsnode.type == "function_item" {  
  let function = Function {  
    name: process_name(tsnode.get_field("name")),  
    type: process_type(tsnode.get_field("return_type")),  
  }  
  process_parameters(tsnode.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

## C++

```
if tsnode.type == "function_definition" {  
  let declarator = tsnode.get_field("declarator");  
  let function = Function {  
    name: process_name(declarator.get_field("declarator")),  
    type: process_type(tsnode.get_field("type")),  
  }  
  process_parameters(declarator.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

## Python

```
if tsnode.type == "function_definition" {  
  let function = Function {  
    name: process_name(tsnode.get_field("name")),  
    type: process_type(tsnode.get_field("return_type")),  
  }  
  process_parameters(tsnode.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

# How it works / CST Traversal / Generalization

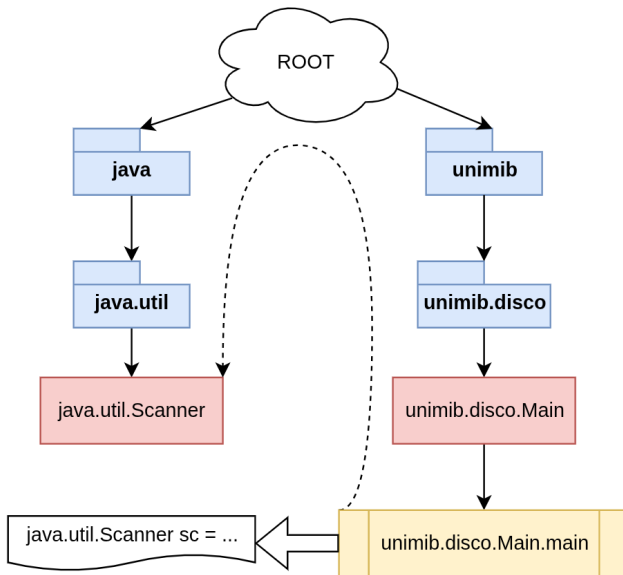
## Java / Rust / Python

```
if tsnode.type == Lang.tags.defun {  
  let function = Function {  
    name: process_name(tsnode.get_field(Lang.tags.defun.direct.name)),  
    type: process_type(tsnode.get_field(Lang.tags.defun.direct.type)),  
  }  
  process_parameters(tsnode.get_field(Lang.tags.defun.direct.parameters));  
  process_body(tsnode.get_field(Lang.tags.defun.direct.body));  
}
```

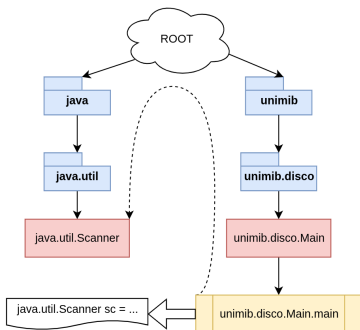
## C / C++

```
if tsnode.type == Lang.tags.defun {  
  let declarator = tsnode.get_field(Lang.tags.defun.declarator.declarator);  
  let function = Function {  
    name: process_name(declarator.get_field(Lang.tags.defun.declarator.name)),  
    type: process_type(tsnode.get_field(Lang.tags.defun.declarator.type)),  
  }  
  process_parameters(declarator.get_field(Lang.tags.defun.declarator.parameters));  
  process_body(tsnode.get_field(Lang.tags.defun.declarator.body));  
}
```

# How it works / Dependency Resolution / Identifier Chains



# How it works / Dependency Resolution / Identifier Chains

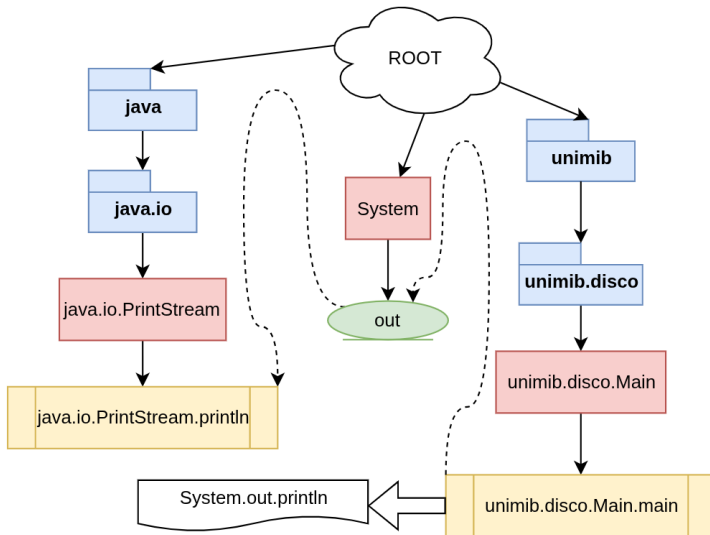


```
def resolve(scope: Node, reference: list[str]):
    assert len(reference) > 0
    scope = find_parent(reference.pop())
    while len(reference) > 0 and scope is not None:
        scope = find_child(scope, reference.pop())
    return scope
```

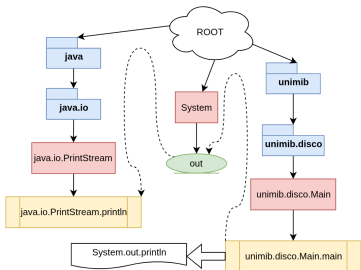
```
def find_parent(scope: Node, identifier: str):
    while scope is not None:
        for child in children(scope):
            if name(child) == identifier:
                return child
        scope = parent(scope)
    return None
```

```
def find_child(scope: Node, identifier: str):
    for child in children(scope):
        if name(child) == identifier:
            return child
    return None
```

# How it works / Dependency Resolution / Type Resolution



# How it works / Dependency Resolution / Type Resolution



```
def resolve(scope: Node, reference: list[str]):
    assert len(reference) > 0
    scope = find_parent(reference.pop())
    while len(reference) > 0 and scope is not None:
        if kind(scope) == FIELD:
            scope = type(scope)
            scope = find_child(scope, reference.pop())
    return scope
```



# How it cheats / Ghost Nodes

## Problem

- Code definitions are caught through CST traversal
- References can be solved iff both source and target are on the graph
- TPL's components cannot be detected because their code isn't read
- References TPL's components cannot be detected because they aren't on the graph

## Example

```
import fi.foyt.foursquare.api.FoursquareApi;
```

## Solution

- Every time we should find a component but nothing is there
- Ex: a method 'foo' inside class 'FoursquareApi' which isn't there
- Create a 'ghost' node with volatile a kind so that we can decide what to do

# How it cheats / Operator Overloading

## Problem

- Some programming languages allow operator overloading
- Syntactically, these are treated as binary/monary operators
- Overloaded operators carry complex logic

## Example

```
void main() {  
    Vector3 A = Vector3 {0,0,1}, B = Vector3 {1,0,0};  
    Vector3 C = A + B; // A.operator+(B)  
}
```

## Solution

- Usually, programming languages define overloading as methods
- We can intercept those method definitions knowing their encoded name
- We can guess their name since they usually have a naming schema
- This data can be added to the language feature set

# How it cheats / Deref Trait

## Problem

- Rust and other languages allow definitions of Delegate Operands
- These delegates are not defined as parent classes or target of implementations
- Languages don't offer clear indicators at component definition level

## Example

```
impl Deref for StructC { type Target = StructA;  
    fn deref(&self) -> &Self::Target { &self.0 }}  
fn main() {  
    StructC::new().method();  
}
```

## Solution

- Usually these are implemented as methods or attributes
- Knowing how they are named, we can intercept them

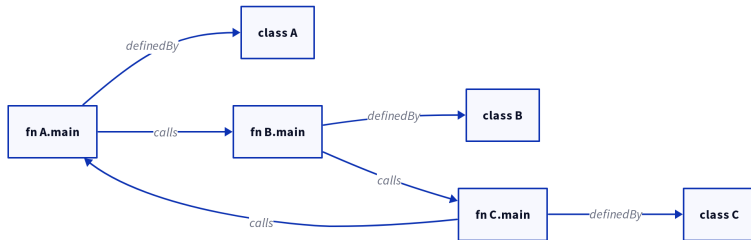
# An Example / The Code

```
class A {  
    public static void main() {  
        B.main();  
    }  
}
```

```
class B {  
    public static void main() {  
        C.main();  
    }  
}
```

```
class C {  
    public static void main() {  
        A.main();  
    }  
}
```

# An Example / The Dependency Graph



# Evaluation and Comparison

# The Testing Framework

Tests are described with YAML manifests including constraints on nodes and edges that must be detected in the dependency graph written by the tool

```
filepaths:
  - main.java
nodes:
  - name: ClassName
    kind: class
  - name: ClassName.MethodName
    kind: function
edges:
  - source: ClassName.MethodName
    sink: ClassName
    kind: definedBy
```

- node types: class, function, enum ...
- edge types: definedBy, includes, usesType, calls ...

# The Dependency Detection Benchmark

| Dependency type             | Example code                                                             |
|-----------------------------|--------------------------------------------------------------------------|
| <b>Import</b>               |                                                                          |
| Class import                | <code>import ModuleB.ModuleB2.Class3;</code>                             |
| <b>Declaration</b>          |                                                                          |
| Instance variable           | <code>private Class3 class3;</code>                                      |
| Class variable              | <code>private static Class3 class3;</code>                               |
| Local variable              | <code>public void method() Class3 class3;</code>                         |
| Parameter                   | <code>public void method(Class3 class3)</code>                           |
| Return type                 | <code>public Class3 method()</code>                                      |
| Exception                   | <code>public void method() throws Class4throw new Class4 ("...");</code> |
| Type cast                   | <code>Object o = (Class3) new Object();</code>                           |
| <b>Call</b>                 |                                                                          |
| Instance method             | <code>variable = class3.method();</code>                                 |
| Instance method-inherited   | <code>variable = class3.methodSuper();</code>                            |
| Class method                | <code>variable = class3.classMethod();</code>                            |
| Constructor                 | <code>new Class3();</code>                                               |
| Inner class                 | <code>method variable = class3.InnerClass.method();</code>               |
| Interface method            | <code>interface I.interfaceMethod();</code>                              |
| Library class method        | <code>libraryClass1.libraryMethod();</code>                              |
| <b>Access</b>               |                                                                          |
| Instance variable           | <code>variable = class3.variable;</code>                                 |
| Instance variable-inherited | <code>variable = class3.variableSuper;</code>                            |
| Class variable              | <code>variable = Class3.classVariable;</code>                            |
| Constant variable           | <code>variable = class3.constantVariable;</code>                         |
| Enumeration                 | <code>System.out.println(Enumeration.VAL1);</code>                       |
| Object reference            | <code>method(class3);</code>                                             |
| <b>Inheritance</b>          |                                                                          |
| Extends class               | <code>public class Class1 extends Class3</code>                          |
| Extends abstract            | <code>class Idem but in this case Class3 should be abstract.</code>      |
| Implements interface        | <code>public class Class1 implements Interface1</code>                   |
| <b>Annotation</b>           |                                                                          |
| Class annotation            | <code>@Class3</code>                                                     |



# The Dependency Detection Benchmark

| Dependency type                | Example code                                                                                           |
|--------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Call</b>                    |                                                                                                        |
| Instance method                | <code>variable = class2.class3.method();</code>                                                        |
| Instance method-inherited      | <code>variable = class2.methodSuper();</code>                                                          |
| Class method                   | <code>variable = class2.class3.classMethod();</code>                                                   |
| <b>Access</b>                  |                                                                                                        |
| Instance variable              | <code>variable = class2.class3.variable;</code>                                                        |
| Instance variable-inherited    | <code>variable = class2.variableSuper();</code>                                                        |
| Class variable                 | <code>variable = class2.class3.classVariable;</code>                                                   |
| Object reference-Reference var | <code>variable = class2.method(class2.class3.class4);</code>                                           |
| Object reference-Return value  | <code>Object o = (Object) class2.getClass4();</code>                                                   |
| <b>Inheritance</b>             |                                                                                                        |
| Extends-implements variations  | <code>public class Class1 extends Class2</code><br><code>public class Class2 extends SuperClass</code> |

1

<sup>1</sup>The accuracy of dependency analysis in static architecture compliance checking, Pruijt et Al, Softw. Pract. Exp. 2017

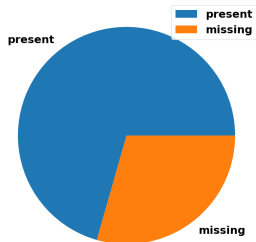


Figure: Adherence of Arcan

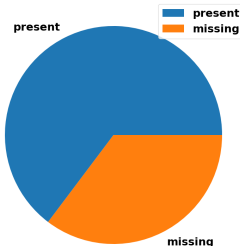


Figure: Adherence of the Stack-Graph based solution

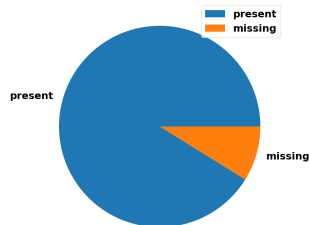
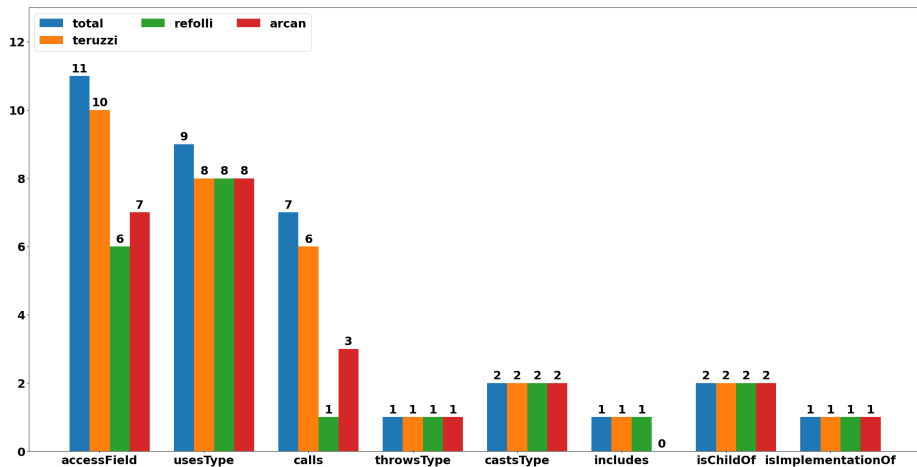


Figure: Adherence of the new proposed solution



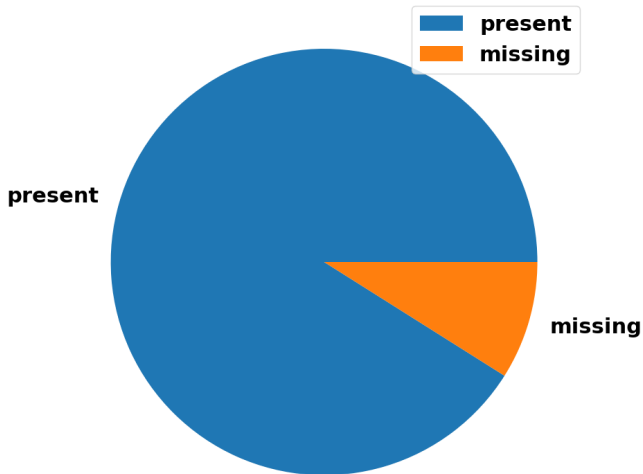
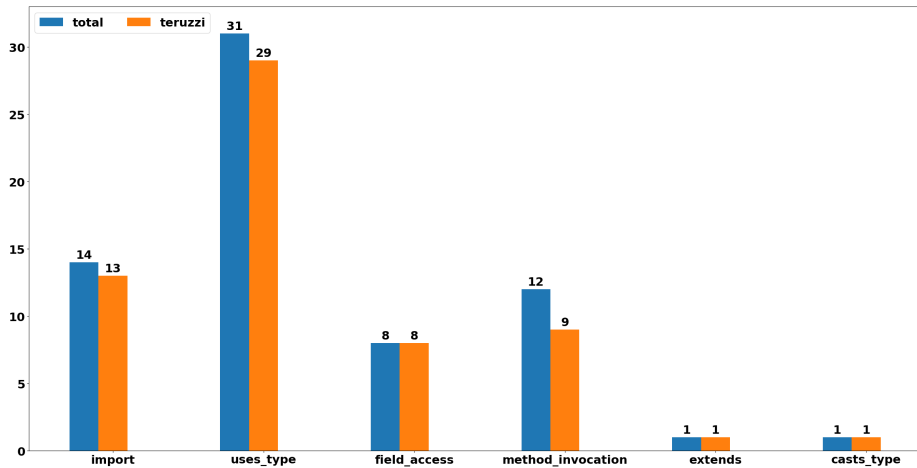


Figure: Adherence of the new proposed solution

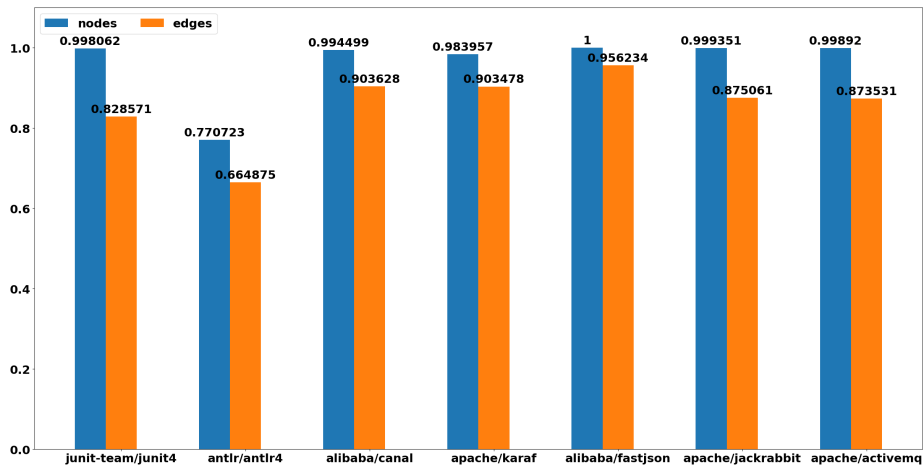


# Comparison with the previous iteration

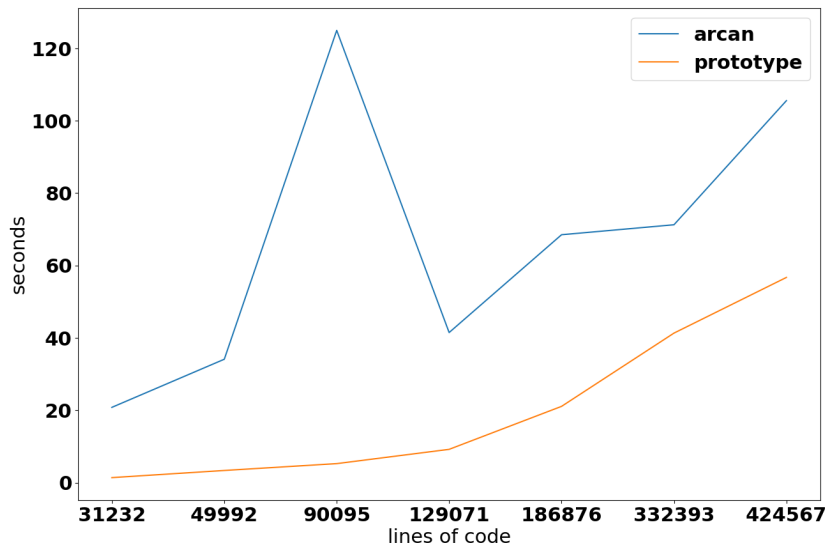
- The Stack-Graph based solution had a decent effectiveness when dealing with small scale projects, but efficiency is another story.
- In some cases it didn't even complete the scan within a reasonable time.
- For these reasons, the comparison is drawn mainly with Arcan in the next slides.

| Project  | Tool        | Min    | Max    | Mean execution time |
|----------|-------------|--------|--------|---------------------|
| JUnit4   | Stack-Graph | 65,82  | 69,07  | 65,88               |
| JUnit4   | Arcan       | 13,850 | 17,079 | 14,611              |
| JUnit5   | Stack-Graph | 132,87 | 134,75 | 134,44              |
| JUnit5   | Arcan       | 42,542 | 47,613 | 44,186              |
| ANTLR    | Stack-Graph | 171,65 | 175,49 | 172,64              |
| ANTLR    | Arcan       | 19,222 | 20,140 | 19,691              |
| Fastjson | Stack-Graph | N/A    | N/A    | N/A                 |
| Fastjson | Arcan       | 66,932 | 71,094 | 69,071              |

# Comparison with Arcan / Similarity



# Comparison with Arcan / Efficiency





# Conclusions

# Open Problems

## Functional Problems

- Package name detection is not working properly
- Multi-crate projects are not supported
- Traits are not fully supported
- Generics are not supported

## Structural Problems

- The prototype is exploratory, with readability issues due to inexperience
- Early work hinders future progress and will need a full rewrite
- Caching and multithreading are missing
- The graph includes debug fields and doesn't handle duplicate dependencies
- File discovery is not properly handled

## Consequences

- The prototype was an exploratory artifact, answering to a specific research question: "is it possible?"
- As such it has accomplished its purpose: "now it can be trashed"
- More work shall be done in order to get an usable framework

## Research Path

- Formalize the approach (and maybe prove its correctness)
- Design an implementation scheme in advance
- Rewrite the prototype into its 3rd iteration

# Thank You

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